

Requirements Engineering for Information Systems

Assignment 2

User stories and features:

a) Family member persona:

Micheal, age 45, living in Portland, is the single father of two 11-years old twin girls. He is a full time hairdresser and has a very busy schedule, and often gets unexpected appointments. Because of this, he wants to improve his communication with his daughters and his household management. Since he isn't using computers a lot for his work, he possesses only very basic skills in IT, as such he values easy to use applications.

Scenario:

Micheal has decided to **improve his and his daughters' organization** by making to do lists, where a member of the household and a time slot are assigned to handle each task. After **having created their to-do tasks** for the upcoming week, he notices that each good to buy in the shopping list is considered an individual task by the application (while the shopping list is a folder grouping the goods).

As this kind of display is too complicated for him, he asks his daughter to **create another display method** for "shopping list" item (a single "shopping list" task with the goods to buy written in its description, still with checkboxes to tick).

→ The application should have many display options i.e. inserting subtasks in the description of a task

b) Company employee persona:

Marina, age 24, living in Hong-Kong, is a graphic designer for a video game company. She uses tools such as Clip Studio Paint to draw her textures and blender to build and animate her models, and finds herself at ease with any digital tools. She is an intuitive person who easily figures how websites and applications work. She is aware of agile practices and sees them as a way to improve her communication with her department

Scenario:

In the context of her project, **Marina** is asked to **report the progress on the current 3D model she is working on**. This report - that she will submit as an attached file to the

“report” task that is shared with her manager in the to-do app- should mention the progress she has done on her assigned tasks and show a preview of her ongoing work. However, when **uploading her 3D model’s components** to the to-do app, she notices it **does not support all file types**, including .blend files. Therefore, to be able to show her mesh, she **uploads it on a cloud and includes a link to it in the report**.

→ The application should support attached files and of many types (excluding potentially malicious types).

c) User stories / Epics for the family:

As a non-expert in the use of IT and applications, I want a tutorial to appear when I install the app, which will explain to me the most important features of the application at first, and continue showing me the details if I choose to try any of them.

As a child, member of a household, I want to create an account on the to-do app using my supervised google account’s credentials.

As a busy parent, I want my children to see my progress percentage on the tasks that I have to do at work (without the details of the task), so they can know when I will be back home.

Time management/Calendar Epic:

{ As a busy parent, I want my children to see my timetable for the upcoming days, and to be able to see theirs, so we can plan activities together that work for both of us.

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As a parent, I want periodic reminders for certain tasks (like going to the hairdresser, getting a control session at the doctor, changing my passport...), so I don’t have to re-enter them each time.

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As a household member, I want to be reminded of the other family members’ birthdays so I don’t forget to buy them a gift. Following the same logic, I want them to automatically be reminded of my birthday, from the moment where I have entered my birthdate when creating my To Do Space account, and I share one or more to-do lists with them. }

Task import/scan/transfer Epic:

{As a daughter in household, I want to have my other family members’ favorite goods being automatically written in the next shopping list, so i don’t have to ask them each time

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As a father who loves DIY tutorials, I want to download tutorials on the internet and import them on the to-do app, so I can easily check the materials I need for my next creation when I go shopping.

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As a daughter who loves to cook, I want to scan recipes I find and get the instructions imported on the to-do app, so I can have them all in one place.}

User stories / Epics for the company employee(s):

As a 3D artist who works depending on certain customer requirements, I want to have the requirements and references I use linked to my task, so I can easily consult them.

As a 3D artist who works mainly on laptop, I want to display my tasks in a 3D view that places them depending on priority, status and type, since it is a view I am comfortable with.

As the artist in my division, I want a comment section under each task, where I can get feedback and eventually likes from my colleagues.

As an employee who often needs to submit reports, I want to easily import my done tasks into the report document that I need to submit to my manager.

As the communicator of my team, I want some of the done tasks to be displayed as teasers on our official social media pages, to keep the target customers' attention high.

As a CEO, I want to use the hierarchy of my company on the to-do app, so the leaders of each division or subdivision can track the progress of their members, and easily communicate with the group they are in charge of.

Additional user stories not related to the previously mentioned scenarios, but from which we can derive features:

As a horse riding instructor, after each training session (a task on the app), I want to write down my trainee's feedback and the horse's mood or behaviour with a small comment and/or an emoji.

I also want to be able to look back at the statistics from the previous sessions to easily track my horses' health and performance.

As an artist sponsored by a company/producer, I want to track the quantity of art supplies I have (with a scrollable/swipeable jar because manually inputting a percentage is too tiring) and get suggestions about new products when those I currently use are bad or empty. I want to be able to select products to buy and have the list sent to my provider automatically. I also want an integrated feedback section on the products that I am using, that pops out when a task is labeled as done/pending/failed...

As an artisan, I want crafting tutorials to be recommended to me depending on the tasks that I have entered (and I want either an AI to sort the tutorials or a feedback function so only the best ones get recommended).

As a scientist or an author, I would like to have an integrated writing space in the app that suggests publication references related to what I am writing. I also want a scientific search engine that allows me to find various sources, and when I use them, I want the sources to

automatically be cited properly at the end of the document, so I don't have to manually create a bibliography.

As a student or novice employee, I want to easily get in touch with people that could help me with the tasks I am entering if I get stuck on them. I want to put subject labels on the tasks, and have subject reference roles for the people I want to contact.

As an eco-conscious child in a not-so-environmental-friendly family, I want my parents to get ecological product suggestions when they type in the name of a good in their shopping list.

d) Features:

-Account creation with Google / Facebook / Org. credentials:

Description: The user may create an account with one of the following methods: An e-mail + password, their Google credentials, their Facebook credentials, or the credentials that their institution has provided if the institution is linked with To Do Space. The method they have used when creating an account will remain the same for future logins.

Constraints: All users must have an account to access To Do Space.

Comments: Consider adding two factor authentication and/or a possibility for the user to log in with several methods.

-Personal information required for the account creation:

Description: Filling basic personal information, like a name, username and birthdate (for legal purposes), should be required to finalize the creation of an account.

Comments: 1. Information like a biography or interest fields tags may be filled in optionally.
2. Data such as the birthdate might be used for other features of the application, like the birthday reminder option on the calendar.

-Account management inside company: hierarchy

Description: The head of an organization or supervisor of a household may create groups (divisions and subdivisions) for the project and give progress monitoring permissions to the leaders of each division / subdivision.

Constraints: Although leaders should be able to monitor their group's work on the project, any employee with a To Do Space account should have the right to create other projects that are kept private if needed, for privacy reasons.

Comments: The progress may be monitored in various ways, for example by using percentages/stats, Kanban boards, reports, etc...

-Tutorial:

Description: The tutorial should appear after the installation of the app, showing the most important features to the user. If the user wants to use more complex features, the tutorial will continue by the moment he tries to use them.

As such, it is a way to find a balance between simplicity and functionality, since the most simple features will be at the foreground, easily accessible to users that seek simplicity, and complex features will still be available for those who seek functionality - but with a little bit of discovery required.

Comment: A “skip tutorial” option should be available to users who are already familiar with this kind of app.

-Dashboard:

Description: All of the user’s ongoing tasks and projects should be visible on their dashboard, the home menu of the To Do *Space*. This is where they will be able to see the folders they created, the recently created tasks, the contact information of the group they belong to, ect...

-Items of the application: tasks, projects, folders, labels, custom items:

Description: Features that are included in the tasks and accounts excluded, the main items of the application should be the tasks, the projects and eventual additional folders inside the project, and the labels assigned to these items. These will serve as a foundation for the different types of display the app will offer.

Comment: The user may have the possibility to create custom items (by choosing a file type that is supported by the app and a custom icon) using the templates of the projects, folders and tasks, if he enables this option.

-Display options for the tasks/project:

Description: *Default display:*

Projects will be displayed as the main/root folders.

Categories will be displayed as columns or lines if a board / table display type is chosen, otherwise they will be displayed as labels. Tasks will be displayed as items within the categories, and will be able to carry several labels.

The location and status of the tasks will be displayed as labels.

Customization: The user should be free to display any items in the form they want it to be (for example, displaying the task statuses as folders). They should also be able to choose different view types such as a Kanban board, a table taking different parameters as columns and lines, an orthonormal coordinate system if there are two parameters, a 3D view if there are 3 parameters, a label/tag list, ect...

Comment: Display type templates should be created to fit the needs of different users without losing the most casual ones into the sea of infinite display possibilities. Companies or organizations should also be able to create their own display type templates and share them to their users.

-Tasks (characteristics):

Description: A task is characterized by a name, path (location in the folders), description, status, labels/tags creation and modification dates, and optional due date.

Constraints: To exist, a task must at least have a name that is unique within the folder it belongs to. Other fields can be left blank if needed.

Comments: There may be different task formats in use if we consider import/export options. Imported tasks of a different format than the default one may have less characteristics.

-Task “issue” status:

Description: In addition to the common “To-Do”, “Ongoing” and “Done” statuses provided by usual to-do applications, To Do Space provides an “issue” status for the tasks, that will automatically send a notification to (or display the contact info of) the project or group manager (or responsible person for technical help) when activated, in order to help the user fixing their problem as quickly as possible.

Constraints: The user may only get helped by members of his team and on open projects he belongs to, which are both defined by his organization. A global collaboration / help feature for all types of projects has not yet been considered as it would require to complexify the structure of the accounts and groups / networks.

Comment: Other custom statuses may be applied to the task if the user wants to.

-Progress tracking (with or without details, private/public view):

Description: A progress tracking feature will be available in different forms to satisfy the needs of different actors. For example, for an external actor, a rough estimation of the project progress through charts may be displayed, as confidentiality on the project content may be needed. For project managers on the other hand, a detailed view of the progress on each task may be required.

This feature may display the progress of a project or task using percentages, charts, colored indicators, remaining work time estimations, or a detailed view of completed / ongoing tasks.

-Task comment section (+emoji feedback):

Description: A comment section may be used for each task or project, where the user can report eventual issues, leave a note to another user, get feedback ect.

Comment: A quick feedback option, using a “like” or an emoji, is to be considered.

- Quicklinks to requirements/guidelines/references:

Description: A user should be able to easily reference another member of his group’s task into, using a quicklink.

Constraints: There are no limits to the kind of task the user can reference as long as they can see it (are a member of the reference’s project group / organization).

Comment: This is only applicable to tasks or projects that contain a description or an attached file, as there is no use in referencing an empty task.

-Attached files function:

Description: It should be possible to attach files to the description tasks.

Constraints: All file types should be supported, except for files that seem to be of a malicious type.

Comment: Depending on the account type a maximum file size to upload may be applied.

-Timetable:

Description: A timetable should be possible to add to the user’s dashboard in order to plan tasks accordingly.

-Calendar for tasks:

Description: For routines or periodically repeated tasks (for example; birthdays, scrum meetings), a calendar may be added to the dashboard. It will be in charge of automatically copying the right tasks at the right moment to the dashboard, following the cycle defined by the user.

Comment: The user should be able to manually stop this process if the routine ends.

-Task content transfer:

Description: Similarly to the calendar, if the task copier detects the creation of a new task with a name that is similar to a previous one (example: Grocery list // Shopping list. The elements that are common in all files having a name containing these words will get copied to the next one that is created).

Constraints: Contrarily to the calendar this duplication process should not be automatic and require the user's approval each time it is suggested to be effective (unless if the user manually sets this feature to automatic).

-Project resources management:

Description: For projects that involve the use of physical resources, the user should be able to track the remaining quantities of each resource, and get usage statistics in order to estimate how long the resources will last. Product suggestions based on the previously used resources, linked with product ratings, might be displayed when the remaining quantities are set to low.

Comment: Here product suggestions can be a source of revenue for the product owner.

-Task import and export:

Description: On the application, the tasks should be saved on a cloud using a specific format, and exportable to the computer using this and/or other formats. The exported tasks should be possible to import back again on any project. As such users would be able to share tasks or projects like tutorials freely as downloadable files on the internet.

Comment: there is a risk that the chosen task format may be used to propagate malware or malicious content on the internet. Prepare countermeasures to this issue during the design of the feature.

Task scan:

Description: An intelligent camera that is able to convert a photo of a simple series of instructions like a recipe or a tutorial to a Project or Task with subtasks should be available.

Comment: This artificial intelligence may require a lot of testing and learning during the phase following the application launch to be functional.

- Social media synchronisation:

Description: If a user links their To Do Space account to their social media account(s) (for example; twitter or facebook), they should be able to publish some of their tasks' content on the linked social media directly within the app.

Constraints: This feature should only be available to the owner or designated responsible party/ies of the project, to avoid any unwanted leaks.

-Intelligent tutorial or product suggestion:

Description: If the title of a task or project corresponds to something recognized by an AI (for example, if the title is “Bake pizza”, or “Buy shampoo”), the AI should suggest related content that could help the user completing his task (here for example, a pizza recipe video, or a shampoo recommendation) directly on the app.

Constraints: To avoid giving irrelevant or even annoying suggestions to the user, the AI should only suggest widely reviewed pages (like popular youtube videos or wikiHow tutorials), at least during the first phase after the application launch. Data from the most used suggestions during this phase could then be collected and used to improve the recommendation algorithm. This feature should also be possible to disable manually by the user.

Comment: Product suggestion can also be used as a way to generate revenue to the product owner.

-Export task to text / report format:

Description: Tasks that are labeled as done should be easily exportable to a visual report template in a text format, so they can be added to the product backlog or sprint backlog if needed.

This template should, by default, show the characteristics of the task (project, category, labels, name, description...), the time spent on it, compared to the estimated time if given (+ additional statistics), the unexpected issues that were faced (if encountered), the comment section of the task (if used), and the final feedback.

Comment: Of course, the user can freely modify this given template in order to get the right information for their product and sprint backlogs.

-Integrated search engine and automatised bibliography:

Description: For some specific tasks like the writing of a text or scientific essay for example, the user should have access to a full text editor linked to the task as the task content/description (or the display of a connected personal text editor like Word, Overleaf, Google docs...). Alongside this text editor should the application be able to display a picture-in-picture integrated search engine (like DuckDuckGo, Google or Ecosia for general needs, or ScienceDirect for scientific needs for example). This combination of an integrated search engine and text editor would then automatise the bibliography, since the workspace directly gathers the references of the publications the author is using. As such, this would provide an all-in-one project workspace for authors and members of the scientific community (hence the name *To-Do Space*).

-Eco-friendly features (to be considered):

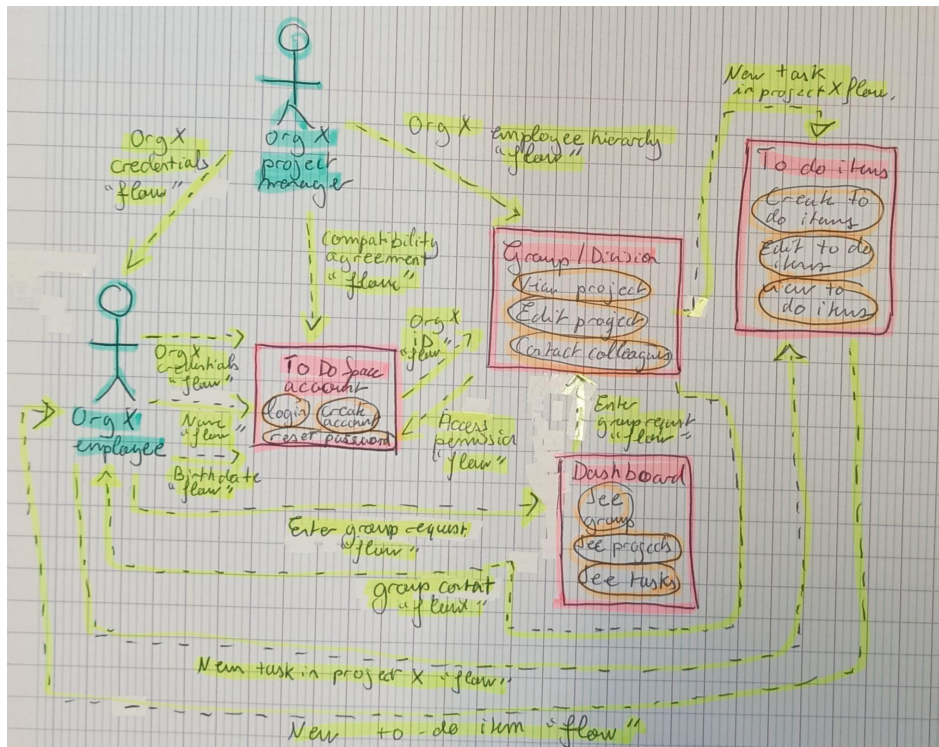
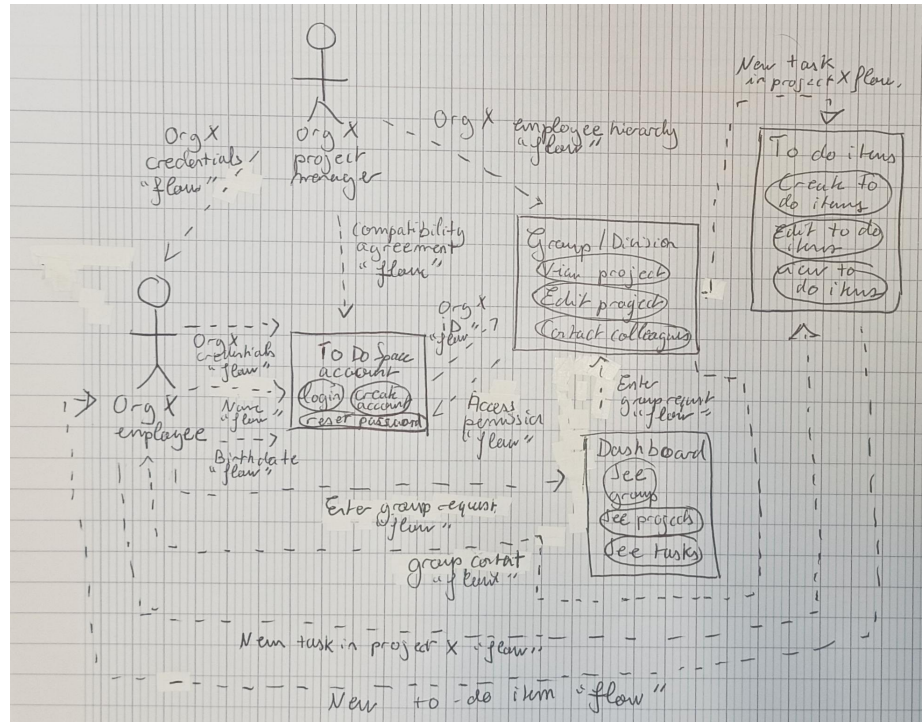
Description: As mentioned in the project resources management, intelligent product suggestions and integrated search engine feature descriptions, external algorithms should support the application to provide these features. This is an opportunity to choose eco-friendly building blocks for these features: like setting Ecosia as a default general search engine, or suggesting environment-friendly products for example.

UML information flows

An overview of the To Do Space application system using the UML notation:

(showing 4 main features: The **account creation with organization credentials**, The **dashboard**, the **Organization account management/Hierarchy** and **tasks**. In other words, the process for an employee to get connected to his organization and start making his first to-do items)

Not Highlighted -



Highlighted for more visibility: with the actors in green, classes in red, use cases in orange and information flows in yellow.